11. Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) breaks the chatbot project into smaller tasks to make the work easier to plan and to manage [3]. This project is divided into six main phases, from planning to completion. Each phase has different tasks that the team needs to complete.

1. Initiation
    - 1.1 Prepare the project charter
    - 1.2 Identify the project stakeholders
    - 1.3 Set the project goals & scope
    - 1.4 Prepare the project management plan
    - 1.5 Get the approval of the project
2. Requirements and Design
    - 2.1 Collect the customer support requirements
    - 2.2 Review previous customer inquiries
    - 2.3 Identify frequently asked questions the chatbot should be able to answer
    - 2.4 Plan how the chatbot will work
    - 2.5 Design the chatbot conversations and knowledge base.
    - 2.6 Get approval for the design
3. Development and Integration
    - 3.1 Set up the chatbot platform
    - 3.2 Develop and train the chatbot to handle the 40 frequently asked questions
    - 3.3 Create the knowledge base
    - 3.4 Connect the chatbot to the existing customer support systems
    - 3.5 Add the chatbot to the website and mobile app
    - 3.6 Create a dashboard to track chatbot performance
4. Testing and Training
    - 4.1 Testing the chatbot responses
    - 4.2 Identifying and resolving errors that come up
    - 4.3 Testing the chatbot security, privacy and performance
    - 4.4 Allow customer support staff to test the chatbot
    - 4.5 Train customer support staff and prepare user's guide
    - 4.6 Obtain final testing approval
5. Deployment
    - 5.1 Complete final check
    - 5.2 Launch the chatbot
    - 5.3 Provide support during the chatbot launch
6. Hypercare and Closure
    - 6.1 Monitor the chatbot
    - 6.2 Resolve issues that comes up after launch
    - 6.3 Gather user and staff feedback
    - 6.4 Prepare the final project report and record lessons learned
    - 6.5 Transfer the chatbot to the IT team
    - 6.6 Close the project

12. Project Task Schedule

The chatbot project will take 24 weeks, start 1st February 2027 and end on 16th July 2027. The table below shows the activities, their start and end dates, how long each task will take, and the person responsible. It also shows which tasks must be completed before others can begin.

Task ID	Project Activity	Start Date	End Date	Duration (Days)	Dependency	Responsible role
1.1	Prepare the project charter	01 February	05 February	5		Project Manager
1.2	Identify the project stakeholders	01 Feb	03 February	3		Business Analyst
1.3	Set the project goals & scope	04 Feb	08 February	3	1.2	Project Manager
1.4	Prepare the project management plan	09 Feb	11 February	3	1.1,1.3	Project Manager
1.5	Get the approval of the project	12 Feb	12 February	1	1.4	Project Sponsor
2.1	Collect the customer support requirements	15 Feb	26 February	10	1.5	Business Analyst
2.2	Review previous customer inquiries	15 Feb	24 February	8	1.5	Business Analyst
2.3	Identify frequently asked questions the chatbot should be able to answer	01 Mar	05 March	5	2.1,2.2	Business Analyst
2.4	Plan how the chatbot will work	01 March	09 March	7	2.1,2.2	Head of IT

2.5	Design the chatbot conversations and knowledge base.	10 March	17 March	6	2.3,2.4	Business Analyst
2.6	Get approval for the design	18 March	19 March	2	2.5	Head of IT
3.1	Set up the chatbot platform	22 Mar	02 April	10	2.6	Head of IT
3.2	Develop and train the chatbot	19 April	07 May	15	3.1,3.3	AI/NLP Engineer
3.3	Create the knowledge base	22 March	16 April	20	2.6	Head of Customer Service
3.4	Connect the chatbot to the existing customer support systems	05 April	07 May	25	3.1	Integration Developer
3.5	Add the chatbot to the website and mobile app	05 April	07 May	25	3.1	Integration Developer
3.6	Create a dashboard to track chatbot performance	10 May	14 May	5	3.4	Integration Developer
4.1	Testing the chatbot responses	17 May	25 May	7	3.2,3.6	QA/ Test Lead
4.2	Identifying and resolving errors that come up	26 May	04 June	8	4.1	AI Engineer

4.3	Testing the chatbot security, privacy and performance	17 May	28 May	10	3.6	QA/ Test Lead
4.4	Allow customer support staff to test the chatbot	07 June	09 June	3	4.2,4.3	Head of customer service
4.5	Train customer support staff and prepare user's guide	31 May	09 June	8	4.3	Head of customer service
4.6	Obtain final testing approval	10 June	11 June	2	4.4,4.5	Head of customer service
5.1	Complete final check	14 June	16 June	3	4.6	Project Manager
5.2	Launch the chatbot	18 June	18 June	1	5.1	Head of IT
5.3	Provide support during the chatbot launch	17 June	18 June	2	5.1	Head of IT
6.1	Monitor the chatbot	21 June	15 July	19	5.2, 5.3	Head of IT
6.2	Resolve issues that comes up after launch	21 June	09 July	15	5.2	AI Engineer
6.3	Gather user and staff feedback	28 June	09 July	10	5.2	Head of customer service
6.4	Prepare the final project report and record lesson learned15	12 July	15 July	4	6.2, 6.3	Project Manager

6.5	Transfer the chatbot to the IT team	12 July	15 July	4	6.2, 6.3	Head of IT
6.6	Close the project	16 July	16 July	1	6.1, 6.4, 6.5	Project Sponsor

Project Milestones

The project has six main milestones. These are the important dates that the team needs to meet before moving to the next stage.

Phase	Milestone	Target Date
Initiation	Charter approved	12 February 2027
Requirements and Design	Design sign off	18 March 2027
Development and Integration	Build complete	14 May 2027
Testing and Training	User Acceptance Testing Sign off	11 June 2027
Deployment	Go-live	18 June 2027
Hypercare and Closure	Project closure	16 July 2027

13. Gantt Chart

The Gnatt chart below shows when each task is expected to start and finish during the 24-week project. It also helps the team see which activities happen at the same time and how the project moves from planning down to completion.

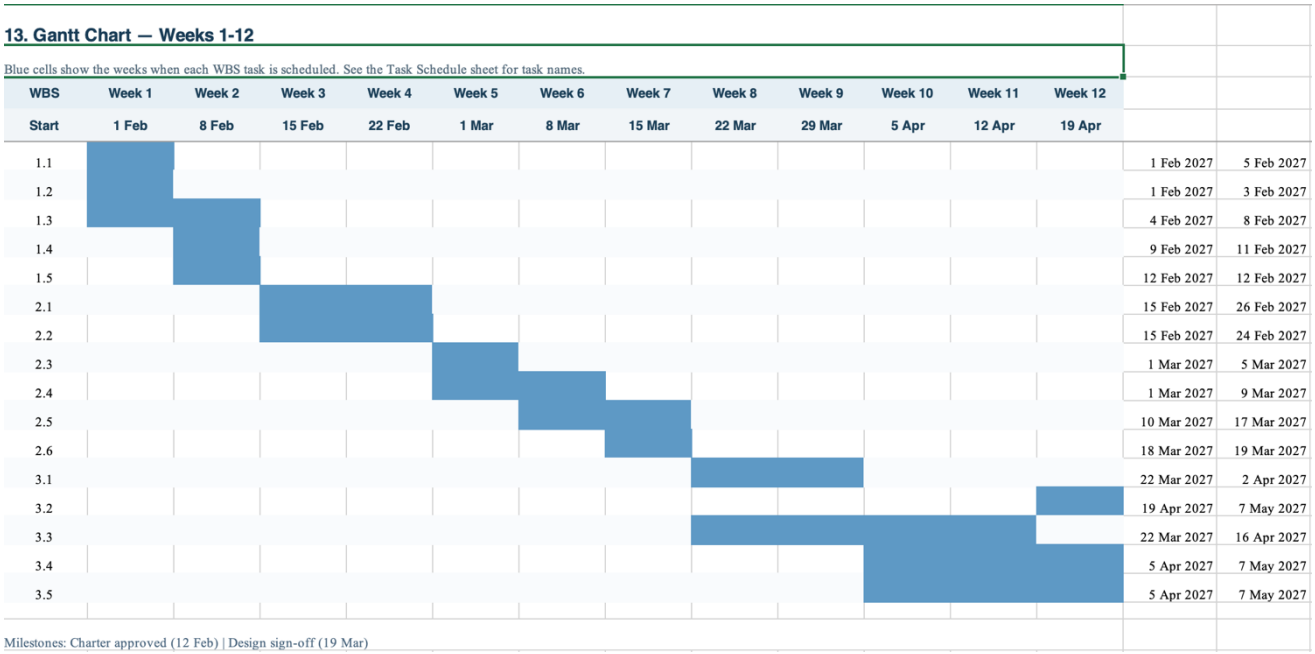


Figure 1: Gantt Chart- Weeks 1-12



Figure 2: Gantt Chart- Weeks 13-24

14. Schedule Explanation

The chatbot project will take 24 weeks, starting on 1 February 2027 and ending on 16 July 2027. The project is divided into six phases, with different tasks planned for each phase.

The project begins with planning and getting approval. After that, the team will collect customer requirements and design the chatbot. Development will begin once the design is approved. Some activities, such as setting up the chatbot platform and preparing the knowledge base, will happen at the same time to save time.

The chatbot will then be tested to make sure it works properly and meets the project requirements. Customer support staff will also test the system and receive training before the chatbot is launched on 18 June 2027.

After the launch, the team will monitor the chatbot for four weeks, fix any problems and collect feedback. The project will officially end on 16 July 2027.

The Gantt chart shows when each task will take place, while the project schedule shows which tasks depend on others. This will help the team track progress and identify any delays that could affect the project.